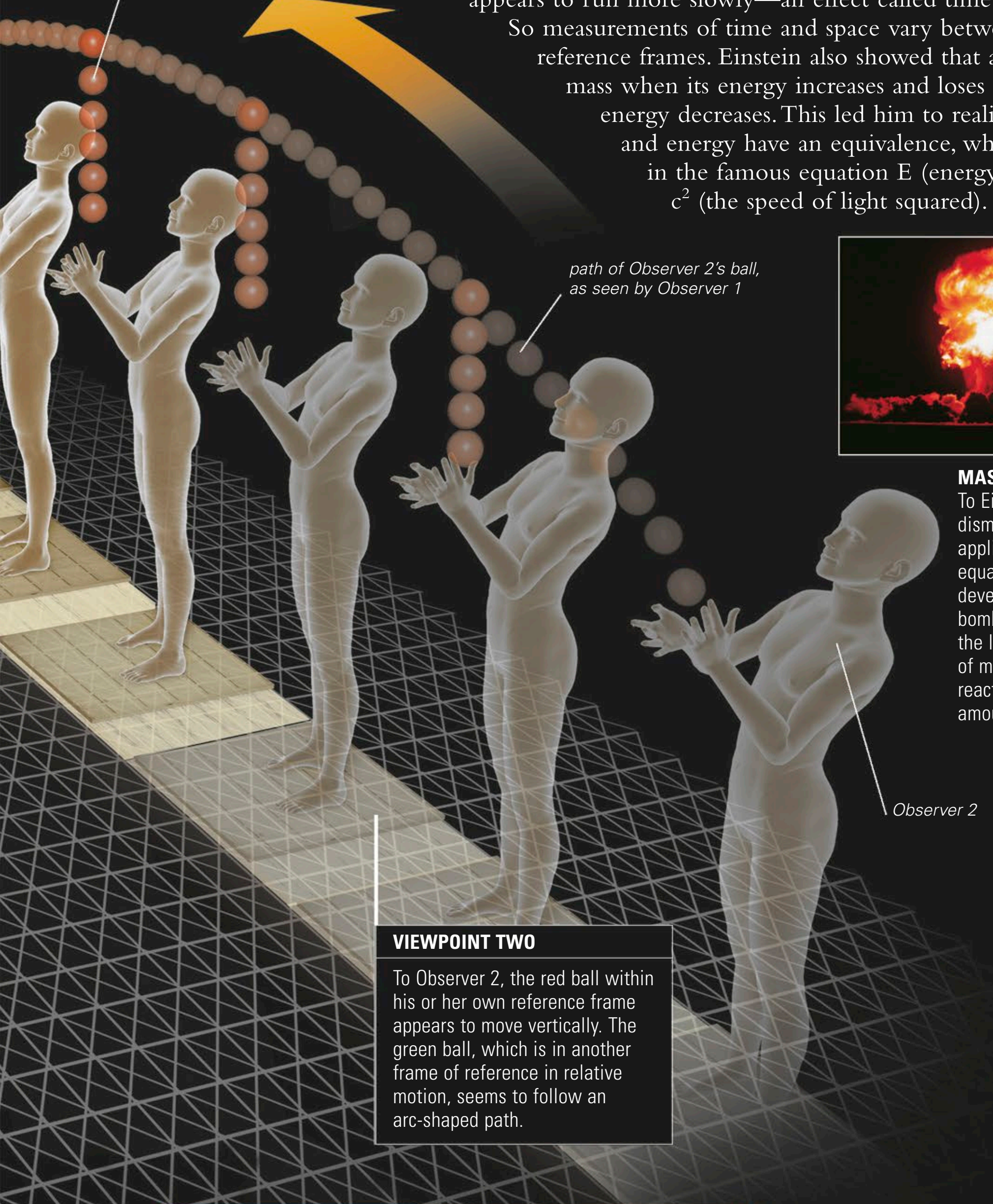
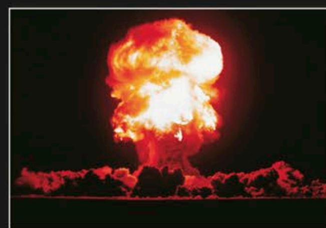
EFFECTS OF SPECIAL RELATIVITY

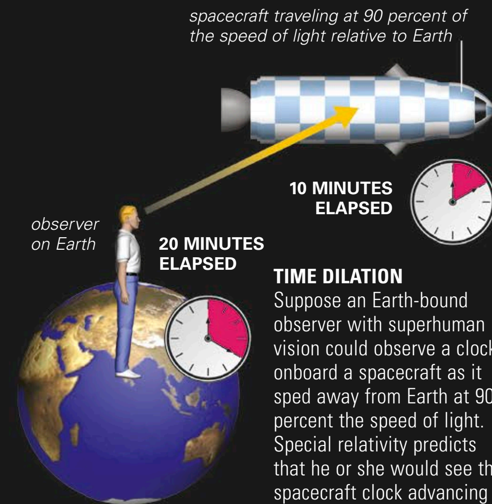
The results that flow from the principles of special relativity are remarkable. Using thought experiments, Einstein showed that for the speed of light to be the same in all reference frames, measures of space and time in one frame must be transformed to those in another. These transformations show that when an object moves at high speed relative to an observer, the observer sees less of its length—an effect called Lorentz contraction. Also, time for such an object appears to run more slowly—an effect called time dilation. So measurements of time and space vary between moving reference frames. Einstein also showed that an object gains mass when its energy increases and loses it when its energy decreases. This led him to realize that mass and energy have an equivalence, which he expressed in the famous equation E (energy) = m (mass) × c^2 (the speed of light squared).



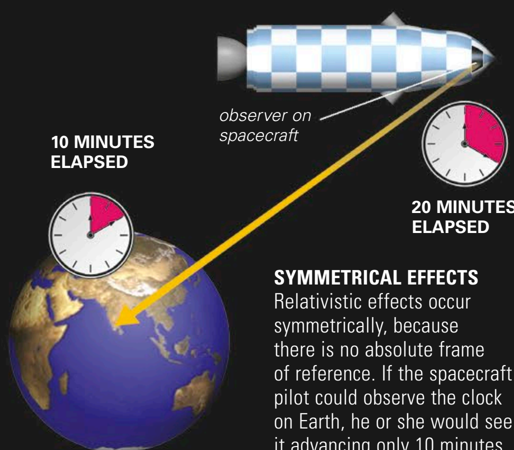
VIEWPOINT TWO
To Observer 2, the red ball within his or her own reference frame appears to move vertically. The green ball, which is in another frame of reference in relative motion, seems to follow an arc-shaped path.



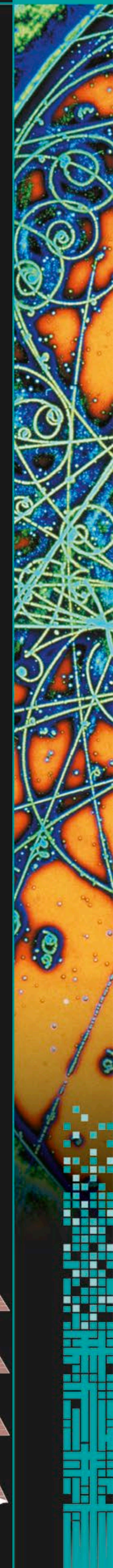
MASS AND ENERGY
To Einstein's ultimate dismay, one of the first applications of his equation $E = mc^2$ was the development of atomic bombs. In such bombs, the loss of tiny amounts of mass in nuclear reactions releases vast amounts of energy.



TIME DILATION
Suppose an Earth-bound observer with superhuman vision could observe a clock onboard a spacecraft as it sped away from Earth at 90 percent the speed of light. Special relativity predicts that he or she would see the spacecraft clock advancing only about 10 minutes while 20 minutes pass on Earth.



SYMMETRICAL EFFECTS
Relativistic effects occur symmetrically, because there is no absolute frame of reference. If the spacecraft pilot could observe the clock on Earth, he or she would see it advancing only 10 minutes while 20 minutes pass on the spacecraft.



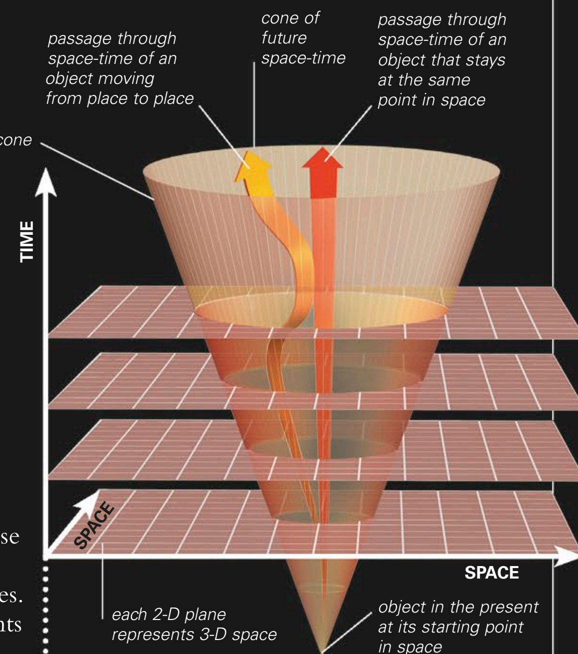
EXPLORING SPACE

PRACTICAL EFFECTS

The prediction from special relativity that time can stretch has been proved to be true by mounting atomic clocks in jet airliners and monitoring their timekeeping compared to Earth-bound clocks. The time dilation effects predicted by special relativity also mean that the atomic clocks in global positioning system (GPS) satellites (below) run a few microseconds a day slower than Earth-bound clocks. General relativity (see p.43) predicts an additional (though reversed) effect, and so data from these orbiting clocks has to be constantly adjusted to maintain accuracy.

SPACE-TIME

A further implication of special relativity is that space and time are closely linked. When two events occur in separate places, the space between them is ambiguous, because observers traveling at different velocities measure different distances. The time passing between the events also depends on each observer's motion. However, a mathematical method can be devised for measuring the separation of events, involving a combination of space and time, that gives values that all observers can agree on. This led to the idea that events in the universe should no longer be described in three spatial dimensions, but rather in a four-dimensional world incorporating time, called space-time. In this system, the separation between any two events is described by a value called a space-time interval.



FOUR DIMENSIONS
In this representation of space-time, time moves upward into the future, while the three spatial dimensions are reduced to two-dimensional planes. The cone represents the effective limits of space-time for any object—its boundary is defined by the speed of light.